

Tutorial sheet - 8

PART - A

Q.1 Define analytical function

⇒ A function $f(z)$ is said to be analytic at a point if it is differentiable in a neighbourhood of that point.

Condition - It must satisfy Cauchy-Riemann (C-R) equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Q.2 Define harmonic function

⇒ A function $u(x, y)$ is called harmonic if it satisfies Laplace equation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Q.3 State conjugate harmonic function.

⇒ If $u(x, y)$ is harmonic, then a function $v(x, y)$ is called its conjugate harmonic function if $f(z) = u + iv$.

Q.4 State Milne Thomson method.

$$f(z) = u + iv$$

(i) If u is given then find $f(z)$

$$\phi_1(x, y) = \frac{\partial u}{\partial x} \quad \text{and} \quad \phi_2(x, y) = \frac{\partial u}{\partial y}$$

Replace x by z and y by 0 , then we get $\phi_1(z, 0)$ & $\phi_2(z, 0)$

$$f(z) = \int [\phi_1(z, 0) - i\phi_2(z, 0)] dz + C$$

(ii) If v is given by $f(z)$ find
 $\psi_2(x,y) = \frac{dv}{dx}$ and $\psi_1(x,y) = \frac{dv}{dy}$

Replace x by z and y by 0 , then we get $\psi_1(z,0)$ and $\psi_2(z,0)$
 $f(z) = \int [\psi_1(z,0) - i\psi_2(z,0)] dz + c$

Q.5 Write Möbius Transformation.

$\Rightarrow$ Mapping upper half-plane to unit circle :- standard result;

$$w = \frac{z-i}{z+i}$$

Properties: Maps $\text{Im}(z) > 0$ to $|w| < 1$

Boundary $\text{Im}(z) = 0$ maps to $|w| = 1$

PART-B

Q.1 Write a short note on conformal mapping.

$\Rightarrow$ A mapping $w = f(z)$ is conformal if it preserves angles and shape locally.
Condition:

- $f(z)$ must be analytic
- $f'(z) \neq 0$

Properties:

- Preserves angle between curves
- Maintains orientation
- Small figures remain similar.

E.g :- $w = z^2$ is conformal except at $z=0$

Q.2 Determine whether $z^{1/2}$ is analytic or not.

$$\Rightarrow f(z) = \frac{1}{z} = \frac{1}{x+iy}$$

Multiply num. and den. by $x-iy$

$$f(z) = \frac{1}{(x+iy)(x-iy)} = \frac{x-iy}{x^2+y^2}$$

$$f(z) = u(x,y) + iv(x,y) = \frac{x}{x^2+y^2} + i \left(\frac{-y}{x^2+y^2} \right)$$

$$u = \frac{x}{x^2+y^2}, \quad v = \frac{-y}{x^2+y^2}$$

$$\frac{du}{dx} = \frac{(x^2+y^2) - x(2x)}{(x^2+y^2)^2}, \quad \frac{du}{dy} = \frac{-x(2y)}{(x^2+y^2)^2}$$

$$\frac{dv}{dx} = \frac{-(x^2+y^2) + y(2y)}{(x^2+y^2)^2}, \quad \frac{dv}{dy} = \frac{-(x^2+y^2) + y(2y)}{(x^2+y^2)^2}$$

$$\frac{du}{dy} = -\frac{dv}{dx} \quad \text{and} \quad \frac{du}{dx} = \frac{dv}{dy}$$

$$x^2+y^2 \neq 0$$

$f(z) = \frac{1}{z}$ is analytic for all $z \neq 0$.

At $z=0$, function is not defined, so not analytic there.

Q. 3 Find the bilinear transformation that maps the point $z_1 = \infty, z_2 = i$ and $z_3 = 0$ into the point $w_1 = \infty, w_2 = i$ and $w_3 = 0$.

$$\Rightarrow z_1 = \infty \rightarrow w_1 = \infty$$

$$z_2 = i \rightarrow w_2 = i$$

$$z_3 = 0 \rightarrow w_3 = 0$$

$$\frac{(w_0 - w_1)(w_2 - w_3)}{(w_1 - w_2)(w_3 - w_0)} = \frac{(z_0 - z_1)(z_2 - z_3)}{(z_1 - z_2)(z_3 - z_0)} \Rightarrow \frac{(w - \infty)(i - 0)}{(\infty - i)(0 - w)} = \frac{(z - \infty)(i - 0)}{(\infty - i)(i - 0)}$$

$$w_1 \left(\frac{w_0}{w_1} - 1 \right) (w_2 - w_3) = z_1 \left(\frac{z_0}{z_1} - 1 \right) (z_2 - z_3)$$

$$w_1 \left(1 - \frac{w_2}{w_1} \right) (w_3 - w_0) = z_1 \left(1 - \frac{z_2}{z_1} \right) (z_3 - z_0)$$

$$\frac{(-1)(i-0)}{(1)(w)} = \frac{(-1)(i-0)}{(1)(0-z)} \Rightarrow \frac{+i}{+w} = \frac{-i}{+z} \Rightarrow w = z \quad \text{Ans}$$

PART - C

Q.1 Find the analytic function $f(z) = u(x,y) + iv(x,y)$ of which the real part $u(x,y) = e^x (x \cos y - y \sin y)$.

$$\Rightarrow u = e^x (x \cos y - y \sin y)$$

$$\phi_1 = \frac{du}{dx} = e^x (\cos y) + e^x (x \cos y - y \sin y), \quad \phi_2 = \frac{du}{dy} = e^x (-x \sin y - \sin y - y \cos y)$$

Replace x by z and y by 0

$$\Rightarrow \phi_1 = e^z (1) + e^z (z) \Rightarrow e^z (1+z), \quad \phi_2 = 0$$

$$f(z) = \int [\phi_1(z,0) + i\phi_2(z,0)] dz + C \Rightarrow \int [e^z (1+z)] dz + C$$

$$= \int e^z dz + \int e^z \cdot z dz \Rightarrow e^z + e^z \cdot z - e^z \Rightarrow z \cdot e^z + C \quad \text{Ans}$$

Q.2 Prove that $f(z) = \sin hz$ is analytic function of the complex variable $z = x + iy$.

$$\Rightarrow \sin hz = \sin h(x + iy) \Rightarrow \sin (x \cos y + i \sin y)$$

$$u = \sin h x \cos y, \quad v = \cos h x \sin y$$

$$\frac{du}{dx} = \cos h x \cos y, \quad \frac{dv}{dy} = \cos y \cos h x$$

First C-R satisfied

$$\frac{du}{dy} = -\sin h x \sin y, \quad \frac{dv}{dx} = -\sin h x \sin y$$

Second C-R satisfied

Since C-R equation hold and functions are continuous $\sinh z$ is analytic everywhere.

Q.3 Find a mobius transformation of the upper half plane $\text{Im}(z) > 0$ on to the interior $|w| < 1$ of the unit circle $|w| = 1$.

$\Rightarrow$ Mobius transformation that converts:

Region: $\text{Im}(z) > 0$ (upper half-plane)

Into: $|w| < 1$ (inside unit circle)

$$w = \frac{z-i}{z+i}$$

A mobius transformation maps: Lines $\leftrightarrow$ circles

Half plane $\leftrightarrow$ Disks

Here:

Real axis (boundary of upper half plane) $\rightarrow$ becomes unit circle $|w| = 1$

$$z = x \text{ (real number, so } \text{Im } z = 0)$$

$$w = \left| \frac{x-i}{x+i} \right| = \left| \frac{x-i}{x+i} \right|, \quad |w| = \frac{-\sqrt{x^2+1}}{\sqrt{x^2+1}} = 1$$

so real axis $\rightarrow$ unit circle

$$z = x + iy, \quad y > 0, \quad w = \frac{z-i}{z+i}$$

$$|w| < 1 \quad \text{if } y > 0$$

so: upper half plane $\rightarrow$ inside circle

Lower half plane $\rightarrow$ outside circle

• Point i goes to $w = \frac{i-i}{i+i} = 0$

∞ center of circle is obtained, point at infinity
 $z \rightarrow \infty \Rightarrow w \rightarrow 1$

$$w = \frac{z-i}{z+i}$$

This transformation maps upper half plane $\text{Im}(z) > 0 \rightarrow$ interior of unit circle $|w| < 1$.